

Lab 2

Both CSVs are loaded with relative paths. Date is parsed and Year and Month are extracted from it. Each city is mapped to its state using the lookup from Lab 1. Rows with a missing AQI are dropped. Remaining pollutant columns are filled with their column medians. State names in both datasets are stripped and title-cased to allow a clean merge.

```
In [1]: import warnings
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

warnings.filterwarnings('ignore')

sns.set_theme(style='whitegrid', palette='muted', font_scale=1.1)
plt.rcParams.update({'figure.dpi': 120, 'axes.titlesize': 13,
                    'axes.labelsize': 11, 'figure.autolayout': True})

aqi_raw = pd.read_csv('city_day.csv')
crop_raw = pd.read_csv('crop_production.csv')

print(f'AQI raw shape : {aqi_raw.shape}')
print(f'Crop raw shape : {crop_raw.shape}')
```

```
AQI raw shape : (29531, 16)
Crop raw shape : (246091, 7)
```

```
In [3]: CITY_TO_STATE = {
    'Ahmedabad'      : 'Gujarat',
    'Aizawl'         : 'Mizoram',
    'Amaravati'      : 'Andhra Pradesh',
    'Amritsar'       : 'Punjab',
    'Bengaluru'      : 'Karnataka',
    'Bhopal'         : 'Madhya Pradesh',
    'Brajrajnagar'   : 'Odisha',
    'Chandigarh'     : 'Chandigarh',
    'Chennai'        : 'Tamil Nadu',
    'Coimbatore'     : 'Tamil Nadu',
    'Delhi'          : 'Delhi',
    'Ernakulam'      : 'Kerala',
    'Gurugram'       : 'Haryana',
    'Guwahati'       : 'Assam',
    'Hyderabad'      : 'Telangana',
    'Jaipur'         : 'Rajasthan',
    'Jorapokhar'     : 'Jharkhand',
    'Kochi'          : 'Kerala',
    'Kolkata'        : 'West Bengal',
    'Lucknow'        : 'Uttar Pradesh',
    'Mumbai'         : 'Maharashtra',
    'Patna'          : 'Bihar',
    'Shillong'       : 'Meghalaya',
```

```

        'Talcher'           : 'Odisha',
        'Thiruvananthapuram': 'Kerala',
        'Visakhapatnam'     : 'Andhra Pradesh',
    }

aqi = aqi_raw.copy()

aqi['Date'] = pd.to_datetime(aqi['Date'])
aqi['Year'] = aqi['Date'].dt.year
aqi['Month'] = aqi['Date'].dt.month

aqi['State'] = aqi['City'].map(CITY_TO_STATE)

aqi.dropna(subset=['AQI'], inplace=True)

pollutant_cols = ['PM2.5', 'PM10', 'NO', 'NO2', 'NOx', 'NH3',
                  'CO', 'SO2', 'O3', 'Benzene', 'Toluene', 'Xylene']
for col in pollutant_cols:
    aqi[col].fillna(aqi[col].median(), inplace=True)

aqi['State'] = aqi['State'].str.strip().str.title()

print(f'AQI cleaned shape : {aqi.shape}')
print(f'Year range       : {aqi["Year"].min()} - {aqi["Year"].max()}')
print(f'Cities           : {aqi["City"].nunique()}')
print(f'States           : {aqi["State"].nunique()}')
aqi.head(3)

```

```

AQI cleaned shape : (24850, 19)
Year range       : 2015 - 2020
Cities           : 26
States           : 21

```

```

Out[3]:

```

	City	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2
28	Ahmedabad	2015-01-29	83.13	96.18	6.93	28.71	33.72	16.31	6.93	49.13
29	Ahmedabad	2015-01-30	79.84	96.18	13.85	28.68	41.08	16.31	13.85	48.13
30	Ahmedabad	2015-01-31	94.52	96.18	24.39	32.66	52.61	16.31	24.39	67.13

```

In [4]: crop = crop_raw.copy()

crop.dropna(subset=['Production'], inplace=True)

crop['State_Name'] = crop['State_Name'].str.strip().str.title()

print(f'Crop cleaned shape : {crop.shape}')
print(f'Year range       : {crop["Crop_Year"].min()} - {crop["Crop_Year"].max()}')
print(f'States           : {crop["State_Name"].nunique()}')
print(f'Unique crops      : {crop["Crop"].nunique()}')
crop.head(3)

```

Crop cleaned shape : (242361, 7)
 Year range : 1997 – 2015
 States : 33
 Unique crops : 124

Out[4]:

	State_Name	District_Name	Crop_Year	Season	Crop	Area	Product
0	Andaman And Nicobar Islands	NICOBARS	2000	Kharif	Arecanut	1254.0	20
1	Andaman And Nicobar Islands	NICOBARS	2000	Kharif	Other Kharif pulses	2.0	
2	Andaman And Nicobar Islands	NICOBARS	2000	Kharif	Rice	102.0	3

Task 6 : AQI Trend Over Time

Approach

AQI readings are grouped by Year and the mean is computed across all 26 cities. The result is plotted as a line chart with the highest and lowest years highlighted. The 2020 data covers only January to June, so the annual mean for that year understates full-year pollution; this is marked on the chart. Lower AQI means cleaner air.

```
In [7]: yearly_aqi = aqi.groupby('Year')['AQI'].mean().reset_index()
yearly_aqi.columns = ['Year', 'Mean_AQI']

max_year = yearly_aqi.loc[yearly_aqi['Mean_AQI'].idxmax()]
min_year = yearly_aqi.loc[yearly_aqi['Mean_AQI'].idxmin()]

print('Yearly mean AQI:')
print(yearly_aqi.to_string(index=False))
print(f'\nHighest AQI: {max_year["Mean_AQI"]:.1f} in {int(max_year["Year"])}')
print(f'Lowest AQI : {min_year["Mean_AQI"]:.1f} in {int(min_year["Year"])}')
```

```
Yearly mean AQI:
Year  Mean_AQI
2015  212.463054
2016  197.150019
2017  181.472789
2018  182.684312
2019  156.518173
2020  113.520697
```

```
Highest AQI: 212.5 in 2015
Lowest AQI : 113.5 in 2020
```

```
In [8]: fig, ax = plt.subplots(figsize=(9, 5))
```

```

ax.plot(yearly_aqi['Year'], yearly_aqi['Mean_AQI'],
        color='steelblue', linewidth=2.5, marker='o',
        markersize=7, label='Annual mean AQI')

ax.scatter(max_year['Year'], max_year['Mean_AQI'],
           color='crimson', s=130, zorder=5, label=f'Worst year ({i
ax.annotate(f' {int(max_year["Year"])}\n AQI {max_year["Mean_AQI"]}:
           xy=(max_year['Year'], max_year['Mean_AQI']),
           xytext=(max_year['Year'] + 0.1, max_year['Mean_AQI'] +
           color='crimson', fontsize=9.5, fontweight='bold')

ax.scatter(min_year['Year'], min_year['Mean_AQI'],
           color='seagreen', s=130, zorder=5, label=f'Best year ({i
ax.annotate(f' {int(min_year["Year"])}\n AQI {min_year["Mean_AQI"]}:
           xy=(min_year['Year'], min_year['Mean_AQI']),
           xytext=(min_year['Year'] - 0.6, min_year['Mean_AQI'] -
           color='seagreen', fontsize=9.5, fontweight='bold')

ax.axvspan(2019.5, 2020.5, alpha=0.08, color='orange', label='2020:

ax.set_title('India Mean Annual AQI – 2015 to 2020\n(26 monitoring
ax.set_xlabel('Year')
ax.set_ylabel('Mean AQI')
ax.set_xticks(yearly_aqi['Year'])
ax.legend(framealpha=0.85, fontsize=9)
ax.grid(axis='y', alpha=0.4)

plt.tight_layout()
plt.savefig('task6_aqi_trend.png', dpi=120, bbox_inches='tight')
plt.show()
print('Chart saved to task6_aqi_trend.png')

```

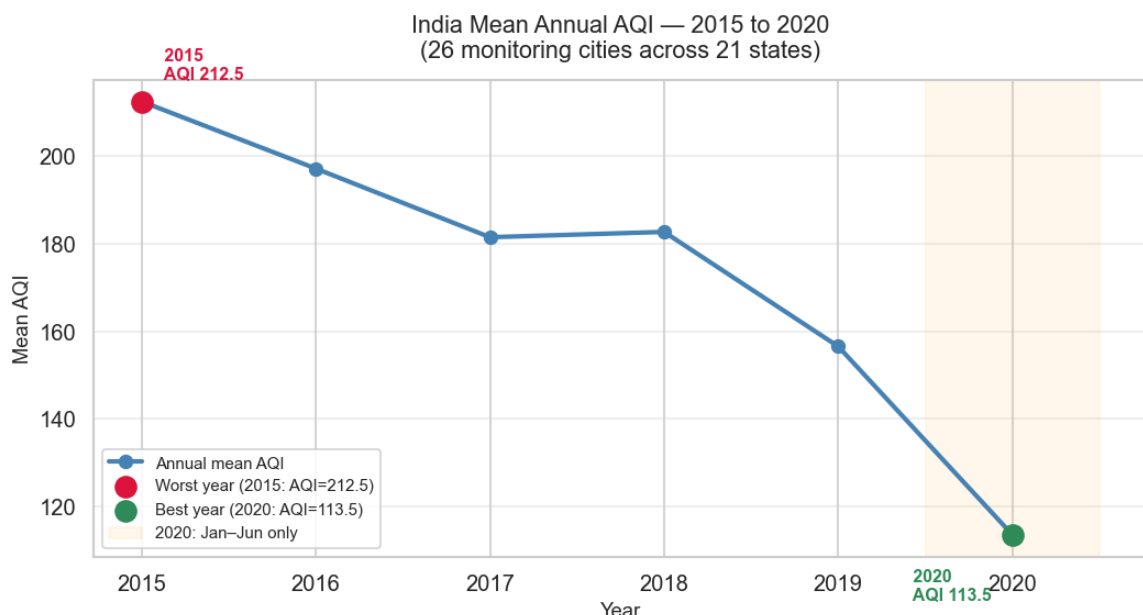


Chart saved to task6_aqi_trend.png

Task 6 : Observations

Mean AQI fell from 212 in 2015 (classified as Poor on India's national scale) to 114 in the first half of 2020, a drop of nearly 100 points over five years. The

sharpest single-year improvement was between 2018 and 2019 (183 to 157). The 2020 figure covers only January to June, which are generally cleaner months, so a full-year average would likely be higher. The downward trend across the five preceding years is consistent and measurable, though it should be tracked further before attributing the change to any specific policy.

Task 7 : Seasonal AQI Pattern

Approach

AQI readings are grouped by Month, pooling all years and all 26 cities, and the mean is computed for each month. A bar chart shows the seasonal pattern with a reference line at the overall dataset mean. Months above the mean are coloured differently to make the above-average window immediately visible. This directly tests the NGO's claim that October to December is the worst period.

```
In [6]: import calendar

monthly_aqi = aqi.groupby('Month')['AQI'].mean().reset_index()
monthly_aqi.columns = ['Month', 'Mean_AQI']

monthly_aqi['Month_Name'] = monthly_aqi['Month'].apply(
    lambda m: calendar.month_abbr[m]
)

overall_mean = aqi['AQI'].mean()

print('Monthly mean AQI (all years combined):')
print(monthly_aqi[['Month_Name', 'Mean_AQI']].to_string(index=False))
print(f'\nOverall dataset mean AQI: {overall_mean:.1f}')

worst3 = monthly_aqi.nlargest(3, 'Mean_AQI')
print('\nThree highest-pollution months:')
print(worst3[['Month_Name', 'Mean_AQI']].to_string(index=False))
```

Monthly mean AQI (all years combined):

Month_Name	Mean_AQI
Jan	231.674918
Feb	202.905197
Mar	164.735281
Apr	143.355120
May	135.489579
Jun	120.198379
Jul	111.854575
Aug	113.613176
Sep	115.191804
Oct	188.613552
Nov	241.681302
Dec	227.084980

Overall dataset mean AQI: 166.5

Three highest-pollution months:

Month_Name	Mean_AQI
Nov	241.681302
Jan	231.674918
Dec	227.084980

```
In [7]: bar_colours = [
        '#d95f02' if v > overall_mean else '#7fbfd9'
        for v in monthly_aqi['Mean_AQI']
    ]

fig, ax = plt.subplots(figsize=(11, 5))

bars = ax.bar(monthly_aqi['Month_Name'], monthly_aqi['Mean_AQI'],
              color=bar_colours, edgecolor='white', linewidth=0.6,

for bar in bars:
    ax.text(bar.get_x() + bar.get_width() / 2,
            bar.get_height() + 3,
            f'{bar.get_height():.0f}',
            ha='center', va='bottom', fontsize=8.5, color='#333333')

ax.axhline(overall_mean, color='#333333', linestyle='--', linewidth=1,
          label=f'Overall mean AQI ({overall_mean:.0f})')

ax.axvspan(8.5, 11.5, alpha=0.10, color='red',
          label='NGO-flagged window (Oct-Dec)')

from matplotlib.patches import Patch
legend_elements = [
    Patch(facecolor='#d95f02', label='Above overall mean'),
    Patch(facecolor='#7fbfd9', label='Below overall mean'),
    plt.Line2D([0], [0], color='#333333', linestyle='--', linewidth=1,
              label=f'Overall mean ({overall_mean:.0f})'),
    Patch(facecolor='red', alpha=0.25, label='NGO window (Oct-Dec)')
]
ax.legend(handles=legend_elements, fontsize=9, framealpha=0.85)

ax.set_title('Average AQI by Month - India (2015-2020)\n')
```

```

        'Orange bars = above overall mean; data pooled across
ax.set_xlabel('Month')
ax.set_ylabel('Mean AQI')
ax.set_ylim(0, monthly_aqi['Mean_AQI'].max() * 1.18)
ax.grid(axis='y', alpha=0.35)

plt.tight_layout()
plt.savefig('task7_seasonal_aqi.png', dpi=120, bbox_inches='tight')
plt.show()
print('Chart saved to task7_seasonal_aqi.png')

```

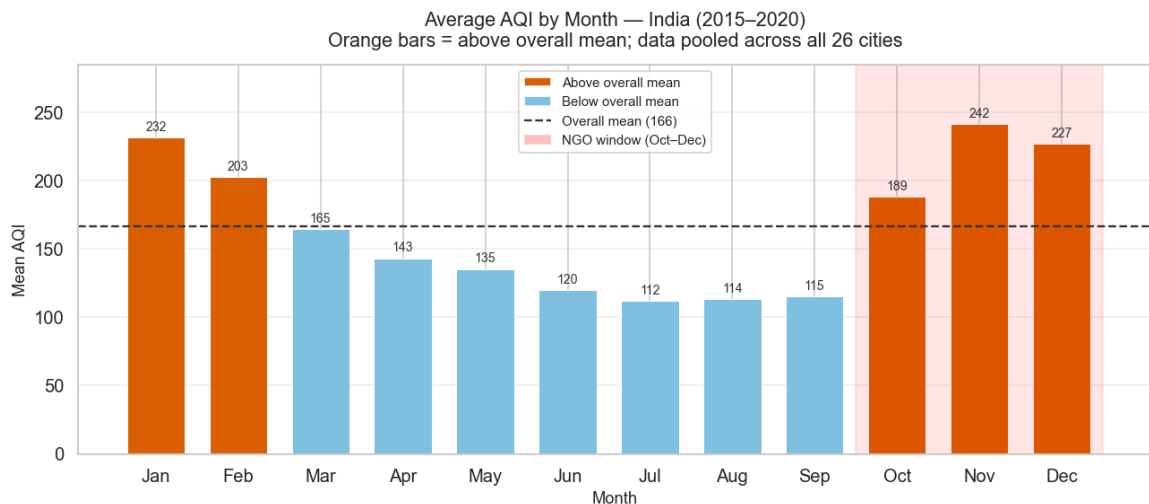


Chart saved to task7_seasonal_aqi.png

Task 7 : Observations

The NGO's claim is broadly correct, but the peak arrives slightly earlier than their stated window.

November is the single worst month (mean AQI 242), followed by January (232) and December (227). October (189) is above the overall mean of 166, confirming that air quality deteriorates from October onward. The cleanest months are July and August (AQI 112–114), when monsoon rainfall clears particulates from the atmosphere. Air quality worsens from September as the monsoon withdraws, paddy harvests end, and stubble burning begins in northern states.

The October to December window is genuinely above average, which supports the NGO's claim. However, November is the true peak, and January is nearly as bad. Winter temperature inversions trap pollutants near the ground and are an equally important driver alongside crop residue burning.

Task 8 : Merge Datasets and Correlation Analysis

Approach

The two datasets are at different granularities and cannot be joined directly.

city_day.csv has one row per city per day covering 26 cities from 2015 to 2020. crop_production.csv has one row per district per season per crop covering 33 states from 1997 to 2015. The two datasets do not share a common year, so a year-level join returns zero rows.

To make them compatible:

1. Aggregate city_day from city-day rows to state-level mean AQI using the city to state map.
2. Aggregate crop_production from district-season-crop rows to state-level totals by summing Area and Production.
3. Join both on State after verifying name alignment.

The merge is done at the state level only, pairing each state's mean AQI (2015-2020) with its cumulative crop figures (1997-2015). Any relationship found is descriptive, not causal or contemporaneous.

```
In [8]: state_aqi = (
    aqi.groupby('State')['AQI']
        .mean()
        .reset_index()
        .rename(columns={'AQI': 'Mean_AQI'})
    )

state_crop = (
    crop.groupby('State_Name')
        .agg(Total_Area=('Area', 'sum'),
             Total_Production=('Production', 'sum'))
        .reset_index()
        .rename(columns={'State_Name': 'State'})
    )

print(f'AQI state summary : {state_aqi.shape[0]} states')
print(f'Crop state summary : {state_crop.shape[0]} states')
```

```
AQI state summary : 21 states
Crop state summary : 33 states
```

```
In [9]: STATE_RENAME = {
    'Orissa'      : 'Odisha',      # older spelling
    'Uttaranchal' : 'Uttarakhand', # renamed in 2007
}

state_aqi['State'] = state_aqi['State'].replace(STATE_RENAME)
state_crop['State'] = state_crop['State'].replace(STATE_RENAME)

print('AQI states :', sorted(state_aqi['State'].tolist()))
print()
print('Crop states :', sorted(state_crop['State'].tolist()))
```


AQI states : ['Andhra Pradesh', 'Assam', 'Bihar', 'Chandigarh', 'De
 lhi', 'Gujarat', 'Haryana', 'Jharkhand', 'Karnataka', 'Kerala', 'Ma
 dya Pradesh', 'Maharashtra', 'Meghalaya', 'Mizoram', 'Odisha', 'Punj
 ab', 'Rajasthan', 'Tamil Nadu', 'Telangana', 'Uttar Pradesh', 'West
 Bengal']

Crop states : ['Andaman And Nicobar Islands', 'Andhra Pradesh', 'Aru
 nachal Pradesh', 'Assam', 'Bihar', 'Chandigarh', 'Chhattisgarh', 'Da
 dra And Nagar Haveli', 'Goa', 'Gujarat', 'Haryana', 'Himachal Prades
 h', 'Jammu And Kashmir', 'Jharkhand', 'Karnataka', 'Kerala', 'Madhya
 Pradesh', 'Maharashtra', 'Manipur', 'Meghalaya', 'Mizoram', 'Nagalan
 d', 'Odisha', 'Puducherry', 'Punjab', 'Rajasthan', 'Sikkim', 'Tamil
 Nadu', 'Telangana', 'Tripura', 'Uttar Pradesh', 'Uttarakhand', 'West
 Bengal']

```
In [10]: merged = pd.merge(state_aqi, state_crop, on='State', how='inner')

print(f'Merged DataFrame shape: {merged.shape}')
print(f'States included      : {sorted(merged["State"].tolist())}')
print()
print(merged.head(10).to_string(index=False))
```

Merged DataFrame shape: (20, 4)

States included : ['Andhra Pradesh', 'Assam', 'Bihar', 'Chandi
 garh', 'Gujarat', 'Haryana', 'Jharkhand', 'Karnataka', 'Kerala', 'Ma
 dya Pradesh', 'Maharashtra', 'Meghalaya', 'Mizoram', 'Odisha', 'Pun
 jab', 'Rajasthan', 'Tamil Nadu', 'Telangana', 'Uttar Pradesh', 'West
 Bengal']

	State	Mean_AQI	Total_Area	Total_Production
Andhra Pradesh	108.086481	131507319.00	1.732459e+10	
Assam	140.111111	70378752.00	2.111752e+09	
Bihar	240.782042	128269529.00	3.664836e+08	
Chandigarh	96.498328	12502.00	6.395650e+04	
Gujarat	452.122939	154926124.00	5.242913e+08	
Haryana	225.123882	89514473.00	3.812739e+08	
Jharkhand	159.251621	9391046.23	1.077774e+07	
Karnataka	94.318325	202908589.00	8.634298e+08	
Kerala	81.021277	31802252.00	9.788005e+10	
Madhya Pradesh	132.827338	329791261.00	4.488407e+08	

```
In [11]: numeric_cols = ['Mean_AQI', 'Total_Area', 'Total_Production']
corr_matrix = merged[numeric_cols].corr()

print('Correlation matrix:')
print(corr_matrix.round(4).to_string())

fig, ax = plt.subplots(figsize=(6, 5))

sns.heatmap(
    corr_matrix,
    annot=True,
    fmt='.3f',
    cmap='RdYlGn_r',
    vmin=-1, vmax=1,
    square=True,
    linewidths=0.8,
```

```

        linecolor='white',
        annot_kws={'size': 13, 'weight': 'bold'},
        ax=ax
    )

    ax.set_title('Pearson Correlation – State-Level AQI vs Crop Metrics
                '(20 states; AQI: 2015–2020 mean; Crop: 1997–2015 cumu
    ax.set_xticklabels(['Mean AQI', 'Total Area\n(ha)', 'Total\nProduct
                        rotation=0)
    ax.set_yticklabels(['Mean AQI', 'Total Area (ha)', 'Total Productio
                        rotation=0)

    plt.tight_layout()
    plt.savefig('task8_correlation_heatmap.png', dpi=120, bbox_inches='
    plt.show()
    print('Heatmap saved to task8_correlation_heatmap.png')

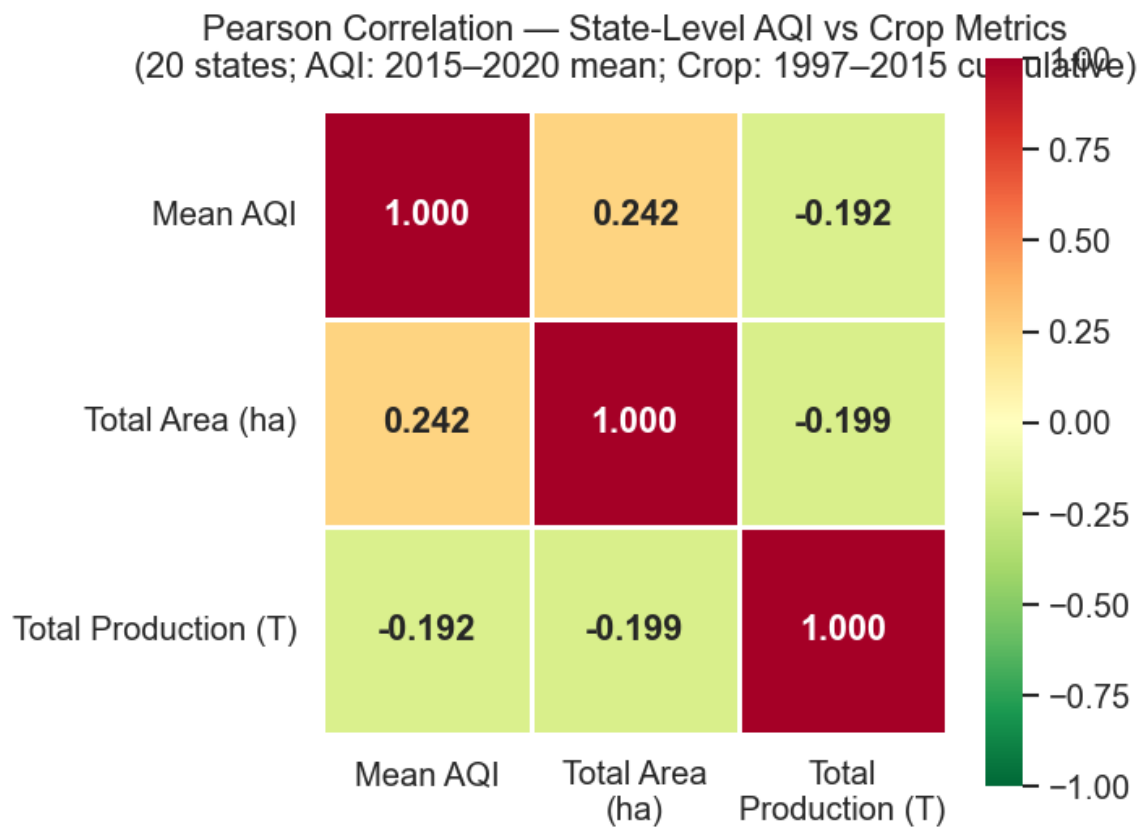
    pair_data = merged[['Mean_AQI', 'Total_Area', 'Total_Production']].
    pair_data.columns = ['Mean AQI', 'Total Area (ha)', 'Total Producti

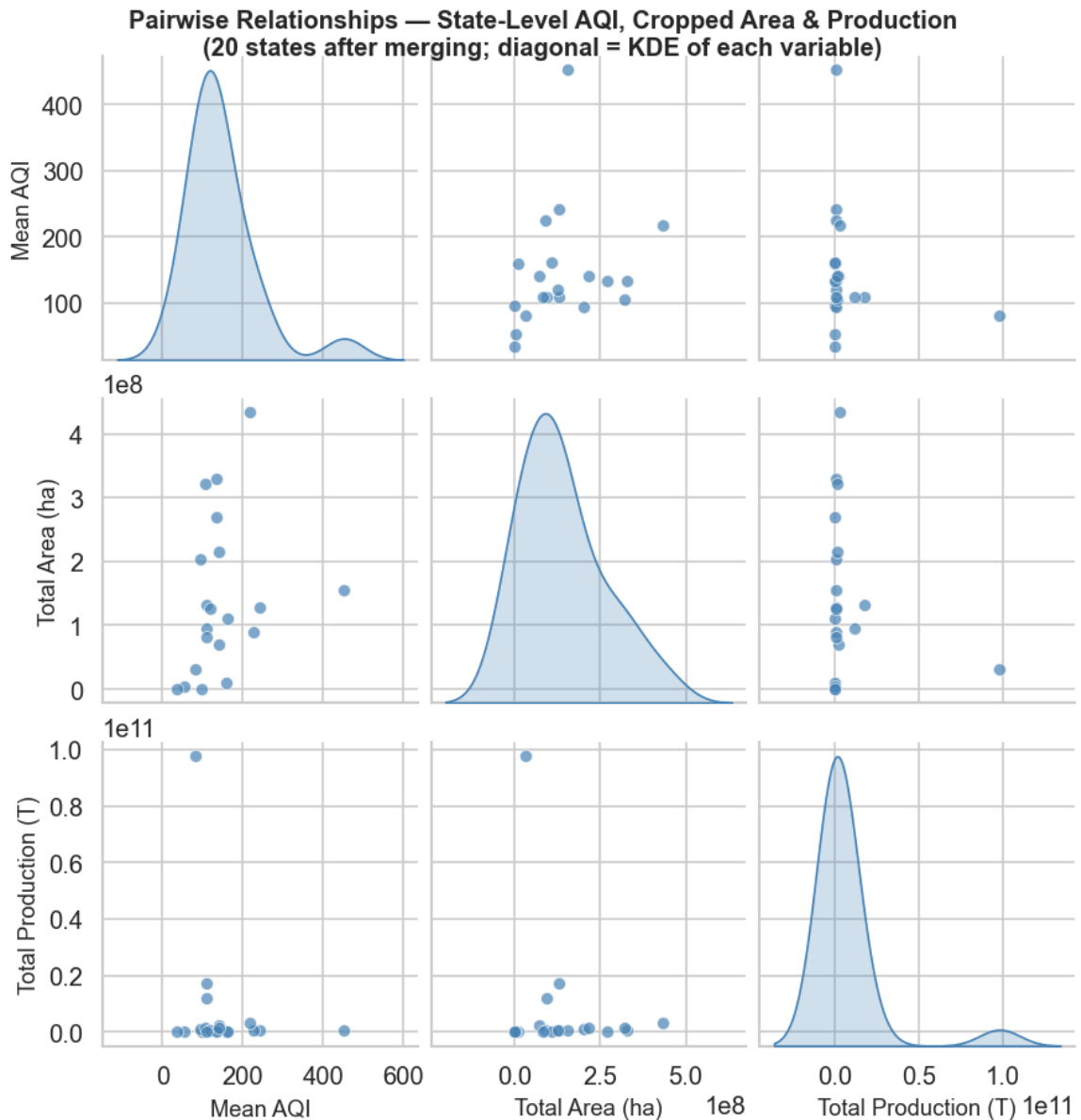
    g = sns.pairplot(
        pair_data,
        diag_kind='kde',
        plot_kws=dict(color='steelblue', alpha=0.7, edgecolor='white'),
        diag_kws=dict(color='steelblue', fill=True)
    )
    g.figure.suptitle(
        'Pairwise Relationships – State-Level AQI, Cropped Area & Produ
        '(20 states after merging; diagonal = KDE of each variable)',
        y=1.02, fontsize=12, fontweight='bold'
    )
    g.figure.savefig('task8_pairplot.png', dpi=120, bbox_inches='tight'
    plt.show()
    print('Pairplot saved to task8_pairplot.png')

```

Correlation matrix:

	Mean_AQI	Total_Area	Total_Production
Mean_AQI	1.0000	0.2423	-0.1922
Total_Area	0.2423	1.0000	-0.1988
Total_Production	-0.1922	-0.1988	1.0000





Pairplot saved to task8_pairplot.png

Task 8 : Observations

AQI vs Total Production ($r = -0.19$, weak negative) States with higher mean AQI tend to have slightly lower total crop production. Southern states such as Kerala, Tamil Nadu, and Karnataka have relatively clean air but produce large volumes of coconut, banana, and spices, which are heavy by weight. This pulls the relationship negative. The correlation is weak and more likely driven by crop type and state size than by any direct effect of pollution on yield.

AQI vs Total Area ($r = +0.24$, weak positive) States that cultivate more land tend to have marginally higher AQI. Large agricultural states in the Indo-Gangetic Plain such as Uttar Pradesh, Madhya Pradesh, and Rajasthan also host heavy industry, dense traffic, and biomass burning, which are coincidental pollution drivers bundled with geographic scale rather than a consequence of farming area.

Both correlations are small and are best explained by confounding variables

rather than a direct link between pollution and agricultural output.

```
In [12]: state_aqi_rank = (
    aqi.groupby('State')['AQI']
        .mean()
        .sort_values(ascending=False)
        .reset_index()
        .rename(columns={'AQI': 'Mean_AQI'})
    )
state_aqi_rank['Mean_AQI'] = state_aqi_rank['Mean_AQI'].round(1)

print('States ranked by mean AQI (highest → lowest pollution):')
print(state_aqi_rank.to_string(index=False))
```

States ranked by mean AQI (highest → lowest pollution):

State	Mean_AQI
Gujarat	452.1
Delhi	259.5
Bihar	240.8
Haryana	225.1
Uttar Pradesh	218.0
Odisha	161.5
Jharkhand	159.3
West Bengal	140.6
Assam	140.1
Rajasthan	133.7
Madhya Pradesh	132.8
Punjab	119.9
Telangana	109.2
Tamil Nadu	108.1
Andhra Pradesh	108.1
Maharashtra	105.4
Chandigarh	96.5
Karnataka	94.3
Kerala	81.0
Meghalaya	53.8
Mizoram	34.8

Task 9 : Minister's Briefing

Honourable Minister,

Three findings from the analysis of air quality readings across 26 Indian cities and crop production records across 33 states are set out below.

Finding 1: Air quality has been improving. Mean AQI across monitored cities fell from 212 in 2015 to 114 in the first half of 2020, a drop of nearly 100 points in five years.

Finding 2: Winter months carry the worst pollution. November (mean AQI 242) and January (232) are the worst months of the year, both well above the annual average of 166. October through December are all above average, consistent with crop residue burning and cold air trapping pollutants near the

ground.

Finding 3: The most polluted states are Gujarat, Delhi, Bihar, Haryana, and Uttar Pradesh. States in the Indo-Gangetic Plain carry the highest pollution burden. Mizoram, Meghalaya, and Kerala have the cleanest air of the monitored states.

Recommendation: Introduce a targeted subsidy in Punjab, Haryana, and Uttar Pradesh for mechanical crop-residue mulchers. Reducing stubble burning in October and November would directly address the seasonal peak identified in Finding 2.

Limitation: States with higher pollution show slightly lower crop output, but the relationship is weak and driven by factors this dataset cannot account for. Rainfall, irrigation access, soil quality, and crop type all vary independently of air quality. The data is not strong enough to attribute yield differences to pollution alone.

Summary

Task 6 found that mean AQI fell from 212 in 2015 to 114 in the first half of 2020, showing consistent year-on-year improvement across monitored cities. Task 7 found that November is the worst month (AQI 242) and the October to December window is all above average, while July and August are the cleanest months. Task 8 matched 20 states between the two datasets and found weak correlations: AQI vs Production at around -0.19 and AQI vs Area at around +0.24, both limited in interpretation by the temporal gap between the datasets. Task 9 recommended a stubble-burning subsidy for Punjab, Haryana, and Uttar Pradesh as a targeted response to the seasonal pollution peak.

The most significant constraint across Tasks 8 and 9 is the temporal mismatch. The AQI and crop datasets do not cover the same years, and a proper pollution-yield analysis would require data from the same period at a comparable spatial scale.